STATE HETEROGENEITY IN

# Australian Retail Macro Response

*A quarterly panel analysis of six states*

**Group 4**

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# Why this project and why this approach

## THE PROBLEM

Australian retail turnover moves with macroeconomic conditions, **but does every state respond the same way?**

## OUR APPROACH

We test whether the same macro shock affects different states differently

## RESEARCH QUESTION

*Do Australian states respond uniformly to macroeconomic shocks, or are there systematic state-specific differences, and do the effects hit immediately or with a lag?*

# From 27 raw workbooks to one clean panel

| 21 retail workbooks | Labour Force | Consumer Prices | Cash Rate | Wage Price | State Final Demand |
|---|---|---|---|---|---|
| *ABS 8501.0* | *ABS 6202.0 Table 12* | *ABS 6401.0 Table 17* | *RBA F1.1 historical* | *ABS 6345.0 Table 1* | *ABS 5206.0 T25* |
| **Turnover** | **Unemployment** | **CPI** | **Policy rate** | **WPI** | **State demand** |

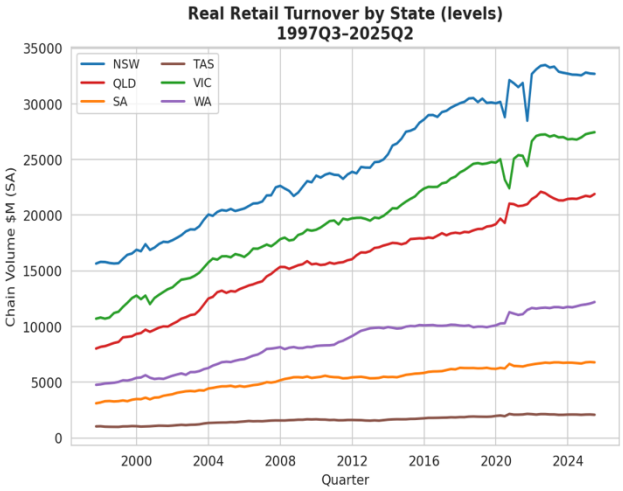
*Python + pandas: standardise state codes → aggregate monthly to quarterly → merge by (Quarter, State)*

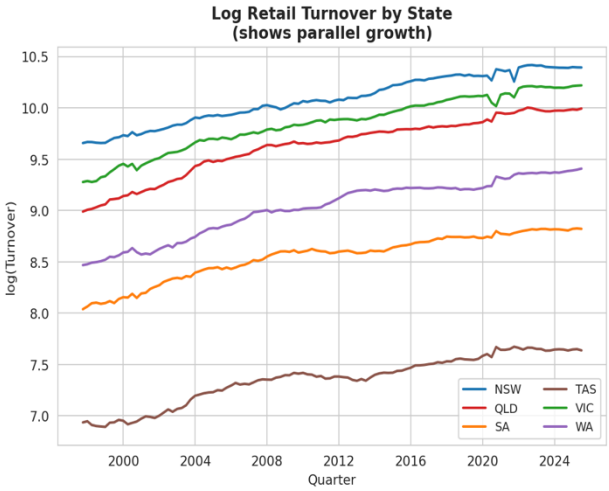
MERGED ANALYSIS PANEL

**672 state-quarter observations** · 6 states × 112 quarters · **1997Q3 – 2025Q2**

*Variables: Retail Turnover (CVM, SA) · Unemployment · CPI · Cash Rate · WPI · State Final Demand*

# State identity dominates the data




## WHAT THIS TELLS US

**NSW is approx. 15× TAS**

Raw-level regressions would be driven by state size, not by real macro effects.

## MODELLING DECISIONS MOTIVATED BY THIS VIEW

**Log-transform the dependent variable · Include state dummies to absorb size differences · Test state × macro interactions**

# Preparing variables for regression

## Log-transformed Y

log(Turnover) converts multiplicative state-size differences into additive ones. Coefficients now read as approximate % effects.

## State dummies

5 dummies encode 6 states (NSW = reference). Let's us test whether each state differs in baseline and response.

## Lag features

groupby('State').shift(1/2) builds 1- and 2-quarter lags without bleeding across states. Tests temporal dynamics.

### MULTICOLLINEARITY CHECK (VIF)

**CPI: VIF = 170 · WPI: VIF = 197 → severe collinearity**

*Both are national trending price indices; they carry redundant information.*

Mitigation: state fixed effects absorb shared national trend; PCA robustness check confirms findings hold.

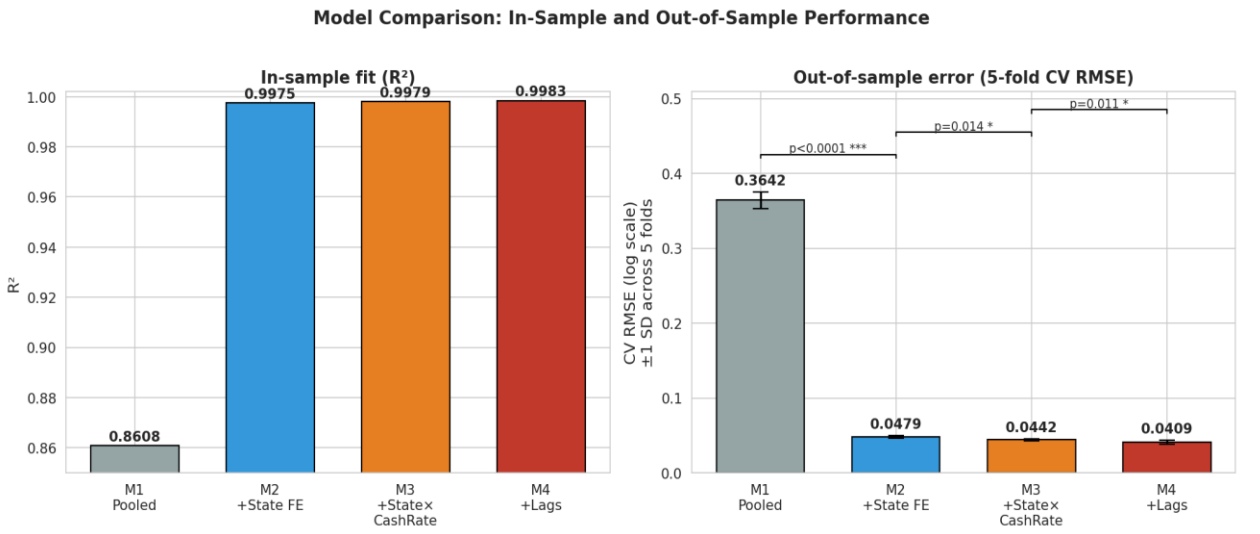
# Four nested regression specifications

| Model | What it adds | Tests this hypothesis |
|---|---|---|
| **M1 — Pooled OLS** | log(Turnover) ~ 5 macro variables | Are macro variables jointly zero? (baseline) |
| **M2 — + State FE** | Adds 5 state dummy variables | Do states have different baselines? |
| **M3 — + Interactions** | Adds 5 State × CashRate interactions | **Do states respond differently to cash rate?** |
| **M4 — + Lags** | Adds 1- and 2-quarter lags of 4 macros | Do macro effects operate with a time lag? |

## STATISTICAL TESTS FOR MODEL COMPARISON

**Nested F-test:** tests whether added predictors are jointly zero (built into statsmodels) · **Paired t-test on CV RMSE:** tests whether complex model predicts better out-of-sample

# Every nested model wins its t-test




**KEY NUMBERS**

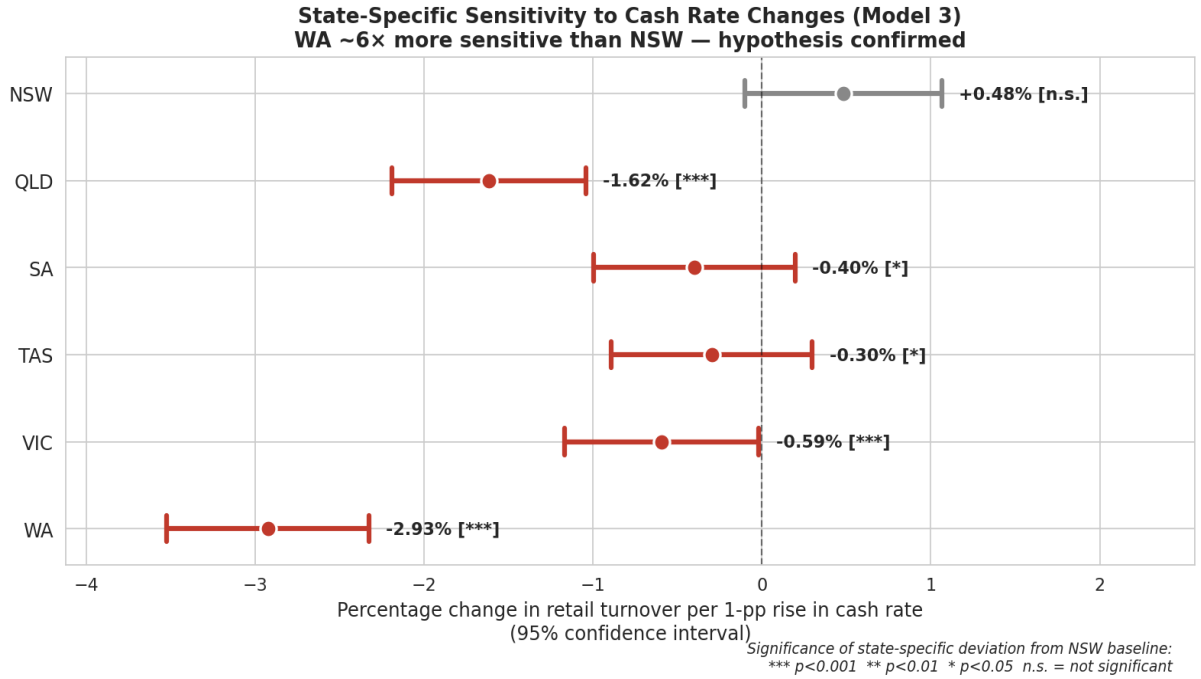
**$R^2$: 0.86 → 0.998**
*when state FE added*

**RMSE drops 87%**
*out-of-sample, M2 vs M1*

**All 3 t-tests reject $H_0$**
*at $p < 0.05$*

*State identity is the dominant source of variation in Australian retail turnover resulting in more than any single macro variable.*

# Cash-rate response differs by state




**WA**

**−2.93%**

*per 1pp cash-rate rise (p < 0.001)*

**NSW**

**+0.48%**

*not statistically significant*

**WA is ~6× more cash-rate sensitive than NSW**

# Why WA differs? theoretical mechanisms

## WESTERN AUSTRALIA

*Why retail is highly rate-sensitive:*

- **Higher average mortgage exposure per household**
- Direct transmission: rate ↑ → mortgage payments ↑ → discretionary spend ↓
- **Resource-sector-linked household income**
- Commodities respond to global credit conditions, amplifying monetary effects

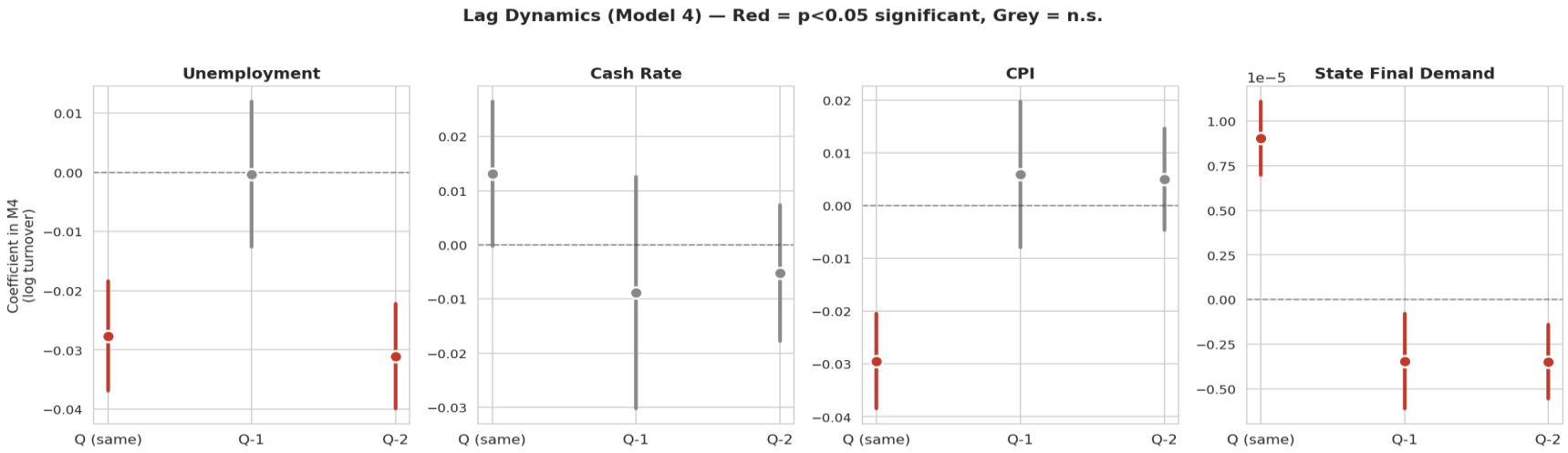
## NEW SOUTH WALES

*Why retail is largely insensitive:*

- **Services / finance-heavy economy**
- Higher rates can boost financial-sector incomes, partially offsetting retail softening
- **Lower household mortgage concentration**
- Diversified rental/owned/investment mix diffuses rate transmission

*Mechanisms are plausible and supported by structural economic evidence — but our regression shows association, not causation.*

# When do macro effects hit retail?




**Unemployment:** Same-quarter effect + persistent 2-quarter lag → cumulative ~6% drag on turnover

**CPI:** Contemporaneous hit only — inflation erodes purchasing power immediately, no overhang

**Cash rate:** Lagged effects not individually significant — the signal lives in state-specific interactions (M3), not in simple temporal delay

# Implications and limitations

## FOR RETAIL DECISION-MAKERS

**State-specific planning:**
WA operations need tighter inventory/staffing response to RBA moves than NSW operations

**Unemployment as early warning:**
Labour-market softening today forecasts retail softening for 2 quarters

**Inflation hits fast:**
No lead-time advantage, but no lingering overhang either

## LIMITATIONS WE ACKNOWLEDGE

**Residual autocorrelation:**
Durbin-Watson ≈ 0.36. Standard errors likely understated; point estimates unbiased

**Association, not causation:**
Regression cannot prove causal channels

**NT & ACT excluded:**
Seasonally-adjusted unemployment data unavailable

CONCLUSION

# Australian retail responds to macroeconomic conditions; but overwhelmingly by state, not by nation.

**~6×**

*WA cash-rate sensitivity vs NSW*

**2 Q**

*Unemployment's persistence horizon*

**0.998**

*$R^2$ with state FE (vs 0.86 pooled)*

**Thank you.** Full analysis and code: *github.com/adnanabiride/PRT564_Assessment-G4*

# Q/A